

Md Musfiqur Rahman Sanim

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SUMMARY

Ph.D. student in Computer Science focused on ML model performance optimization focused for edge devices, with software development experience, seeking a summer 2027 internship.

PUBLICATIONS

Md Musfiqur Rahman Sanim, Zhihao Shu, Bahram Afsharmanesh, AmirAli Mirian, Wei Niu, Gagan Agrawal (2026). “Mask-Aware Execution for Efficient JEPA Training”. In: *Proceedings of the International Conference on Parallel Architectures and Compilation Techniques*. PACT '26. Chicago, Illinois, USA

Summary: An optimized JEPA training framework that exploits cross-mask redundancy through shared execution, fused mask-aware routing, and sparse computation, achieving up to $1.7\times$ faster end-to-end training while reducing redundant computation.

Md Musfiqur Rahman Sanim, Zhihao Shu, Bahram Afsharmanesh, AmirAli Mirian, Jiexiong Guan, Wei Niu, Bin Ren, Gagan Agrawal (2025). “Optimizing 3D Gaussian Splattering for Mobile GPUs”. In: *Proceedings of the International Conference on Parallel Architectures and Compilation Techniques*. PACT '25. Irvine, CA, USA. URL: <https://doi.org/10.1109/PACT65351.2025.00040>

Summary: An optimized 3DGS framework for mobile GPUs that leverages a novel sorting algorithm and memory-aware data placement to efficiently utilize the 2D texture cache, achieving up to $4.1\times$ faster sorting and $1.7\times$ faster overall 3D scene reconstruction with reduced memory usage.

Wei Niu, **Md Musfiqur Rahman Sanim**, Zhihao Shu, Jiexiong Guan, Xipeng Shen, Miao Yin, Gagan Agrawal, Bin Ren (2024). “SmartMem: Layout Transformation Elimination and Adaptation for Efficient DNN Execution on Mobile”. In: *Proceedings of the 29th ACM International Conference on Architectural Support for Programming Languages and Operating Systems*. ASPLOS '24. La Jolla, CA, USA. URL: <https://doi.org/10.1145/3620666.3651384>

Summary: A framework for mobile DNNs that eliminates costly layout transformations by unifying tensor layouts across operators, achieving up to $7.9\times$ speedup over existing frameworks on mobile devices.

WORK EXPERIENCE

Graduate Research Assistant, University of Georgia

Aug 2023 — current

- Conduct research on machine learning systems and GPU optimization, designing experiments and implementing performance improvements for deep learning workloads.

Summer Research Intern, Qualcomm

May 2026 — Aug 2026

- Conduct research on throughput(token/sec.) vs efficiency(token /joule) for running LLM on mobile SoC.

Software Engineer, Enosis Solutions, Dhaka, Bangladesh

Feb 2020 — July 2023

- R&D for mobile and backend systems, deploying large-scale applications on AWS using Docker.
- Designed ML-powered features for production applications, bridging applied research and engineering.

EDUCATION

08/23 - 12/27(expected) PhD in Computer Science at **University of Georgia, Athens, GA, USA**
(GPA: 3.96/4.0)

SKILLS

ML/Systems	CUDA, OpenCL, Vulkan, PyTorch, TensorFlow, Scikit-learn, MNN, TNN
Languages	C++, Python, Java, JavaScript/TypeScript, Swift, Kotlin
Distributed Systems	Node.js, Express, Spring, Django
Databases/Cloud	MySQL, PostgreSQL, AWS, Docker